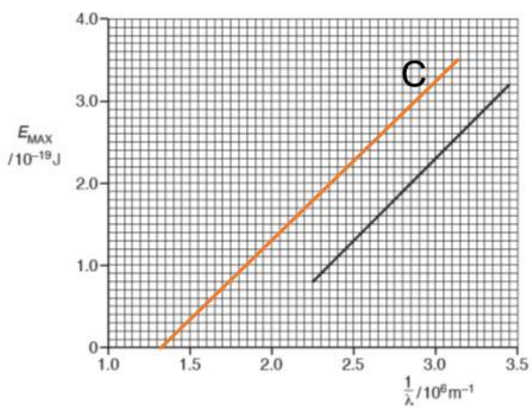
7	**(a)**	minimum frequency of e.m. radiation/a photon (not "light") for emission of electrons from a metal surface	B1
	(b)	E_{MAX} corresponds to electron emitted from surface electron (below surface) requires energy to bring it to surface,so less than E_{MAX}	B1 B1
	(c)(i)	$1/\lambda_0 = 1.85 \times 10^6$ (allow 1.82 to 1.88) $f_0 = c / \lambda_0 = 3.00 \times 10^8 \times 1.85 \times 10^6$ $= 5.55 \times 10^{14}$ Hz	C1 A1
	(c)(ii)	$\Phi = hf_0$ $= 6.63 \times 10^{-34} \times 5.55 \times 10^{14}$ (allow ECF from (c)(i)) $= 3.68 \times 10^{-19}$ J	 C1 A1
	(c)(iii)	The classical wave theory cannot explain the existence of the threshold frequency [since it predicts that if sufficiently intense light is used, the electrons would absorb enough energy to escape.]	B1
	(c)(iv)	No change in the graph [Max kinetic energy is independent of the intensity of light]	B1
	(c)(v)	sketch: straight line with same gradient x- intercept is less since threshold frequency is less  	M1 A1

Section B

8	**(a)**	**(i)**	Amount of substance	B1